

C program

The screenshot shows a C program in a text editor and its execution output. The program, named `binary to decimal.cpp`, uses a while loop to convert a decimal number to binary by repeatedly dividing by 2 and storing the remainders. The output window shows the input 1000, the intermediate remainder 8, and the final binary result 1111101000.

```
1 #include <stdio.h>
2 int main(){
3     int n, decimal=0, base=1, rem;
4     scanf("%d", &n);
5     while(n>0){
6         rem=n%10;
7         decimal=decimal+rem*base;
8         base=base*2;
9         n=n/10;
10    }
11    printf("%d", decimal);
12    return 0;
13 }
```

```
C:\os\binary to decima x + - □ x
1000
8
-----
Process exited after 4.443 seconds with return
value 0
Press any key to continue . . . |
```

The screenshot shows a C program in a text editor and its execution output. The program, named `binary to hexadecimal.cpp`, converts a binary number to a decimal value and then to a hexadecimal string. The output window shows the input binary number 1001, the calculated decimal value 9, and the final hexadecimal result 9.

```
1 #include <stdio.h>
2 int main() {
3     int decimal = 0, base = 1, rem;
4     printf(" binary number: ");
5     scanf("%d", &binary);
6     while (binary > 0) {
7         rem = binary % 10;
8         decimal += rem * base;
9         base *= 2;
10        binary /= 10;
11    }
12    printf("Hexadecimal number = %x", decimal);
13    return 0;
14 }
```

```
C:\os\binary to hexade x + - □ x
Enter binary number: 1001
Hexadecimal number = 9
-----
Process exited after 15.3 seconds with return
value 0
Press any key to continue . . . |
```

```
1 #include <stdio.h>
2 int main(){
3     int n,binary[20],i=0;
4     scanf("%d",&n);
5     while(n>0){
6         binary[i]=n%2;
7         n=n/2;
8         i++;
9     }
10    for (i=i-1;i>=0;i--)
11        printf("%d",binary[i]);
12    return 0;
13 }
```

C:\os\decimal to binary x + - □ x

```
13
1101
-----
Process exited after 2.78 seconds with return
value 0
Press any key to continue . . . |
```

```
1 #include <stdio.h>
2 int main() {
3     int n, rem, octal = 0, place = 1;
4     printf("Enter decimal number: ");
5     scanf("%d", &n);
6     while (n != 0) {
7         rem = n % 8;
8         octal = octal + rem * place;
9         n = n / 8;
10        place = place * 10;
11    }
12    printf("Octal number = %d", octal);
13    return 0;
14 }
```

C:\os\decimal to octal x + - □ x

```
Enter decimal number: 24
Octal number = 30
-----
Process exited after 3.269 seconds with return
value 0
Press any key to continue . . . |
```

The screenshot shows a C++ IDE with the file 'binary to octal.cpp' open. The code implements a function to convert a binary number to an octal number. The main function prompts the user to enter a binary number, which is then converted to octal using a series of division and multiplication steps. The output window shows the execution results for the input '1100', resulting in the octal number '14'.

```
1 #include <stdio.h>
2 int main() {
3     int octal=0, decimal=0, base=1, rem;
4     printf(" binary number: ");
5     scanf("%d", &binary);
6     while(binary > 0){
7         rem = binary % 10;
8         decimal += rem * base;
9         base *= 2;
10        binary /= 10;
11    }
12    base = 1;
13    while(decimal > 0){
14        rem = decimal % 8;
15        octal += rem * base;
16        base *= 10;
17        decimal /= 8;
18    }
19    printf("Octal number = %d", octal);
20    return 0;
21 }
```

Execution Output:

```
Enter binary number: 1100
Octal number = 14
-----
Process exited after 2.629 seconds with return value 0
Press any key to continue . . . |
```

The screenshot shows a C++ IDE with the file 'decimal to hexadecimal.cpp' open. The code implements a function to convert a decimal number to a hexadecimal number. The main function prompts the user to enter a decimal number, which is then converted to hexadecimal using a series of division and remainder operations. The output window shows the execution results for the input '14', resulting in the hexadecimal number 'E'.

```
1 #include <stdio.h>
2 int main() {
3     int n, rem;
4     char hex[20];
5     int i = 0;
6     printf("Enter decimal number: ");
7     scanf("%d", &n);
8     while (n != 0) {
9         rem = n % 16;
10        if (rem < 10)
11            hex[i] = rem + 48;
12        else
13            hex[i] = rem + 55;
14        n = n / 16;
15        i++;
16    }
17    printf("Hexadecimal = ");
18    while (i > 0) {
19        printf("%c", hex[i - 1]);
20        i--;
21    }
22    return 0;
23 }
```

Execution Output:

```
Enter decimal number: 14
Hexadecimal = E
-----
Process exited after 3.167 seconds with return value 0
Press any key to continue . . . |
```